





Dirac or Majorana

- ★ Dirac fermion when **fermion** $\neq$ **anti-fermion**.

Example: electron and positron are Dirac fermions because of opposite charge

- ★ Majorana fermion when fermion = anti-fermion.

Example: neutrino and anti-neutrino could be the same particle

Majorana

Example: neutrino and anti-neutrino could be different particles

Dirac

We do not know if neutrino is a Dirac or Majorana fermion

$$0\nu\beta\beta \Rightarrow \text{Majorana}$$

Neutrino Mass